The constrained flow on the kinetic energy term, on the other hand, is generated by the equations of motion

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}t} &= \boldsymbol{M}^{-1}\boldsymbol{p},\\ \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} &= -\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda},\\ 0 &= \boldsymbol{g}(\boldsymbol{q}).\end{aligned}$$

This is nothing other than the mass-weighted geodesic flow on the given constraint surface. In most cases, the complication of solving the geodesic flow would lead to a need for additional approximations, however it is worth noting that in many applications, by far the most important cost is the calculation of the force $-\nabla U$, hence a further approximation of the geodesic flow could be justified if this method conferred additional stability or other benefits. We do not consider this further here, but take up the construction as part of our treatment of constrained stochastic dynamics in Section 7.12.

For an alternative approach to the construction of higher-order composition methods for constrained systems see [309].

4.3.6 Algorithms for Molecular Simulation Using Constraints

Because of their importance in the molecular simulation process, a wide variety of methods for handling holonomic constraints have been proposed in the past few decades. We summarize a few of these here.

The term "SHAKE" normally refers to both the discretization mentioned above and the component-wise solution of the nonlinear equations (these methods were suggested together in the paper [322]). Using the traditional SHAKE approach we solve the Eqs. (4.27)–(4.29) using the iterative procedure of Sect. 4.3.3. If RATTLE is used for discretization, the iterative method of Sect. 4.3.3 can again be used to solve (4.30)–(4.32), followed by a subsequent solve of Eqs. (4.33)–(4.34).

While SHAKE/RATTLE is a robust procedure the sequential, iterative character of nonlinear solvers needed for its implementation can lead inefficiencies when it is implemented on a parallel computer. Therefore, a number of the alternative methods avoid iteration. We briefly summarize a few of the popular alternative schemes here.

M-Shake This is Newton-iteration-based implementation of SHAKE, using (4.18)–(4.19) to solve Eqs. (4.27)–(4.29). Methods like this were first proposed by Ciccotti and Ryckaert [84] in the context of rigid body molecular dynamics. An extended discussion of such methods with reference to their convergence, implementation, in particular linear system solvers, and variants such as SHAKE-SOR (which uses the "successive over-relaxation" method) can be found in [25]. A conjugate gradient method can also be used [392].